MODULE 8 — Data Science Readiness for Production

Knowing When a System Is Ready (and When It Isn't)

Why This Module Matters

Most data science failures happen **between "it works" and "it's live."**
Teams ship systems that:

- look good in notebooks
- pass technical checks
- satisfy metrics

...but are **not ready to make real decisions**.
This module teaches you to recognize the difference between:
a functioning pipeline and **a decision system safe to operationalize**

Core Reframe (Memorize This)

Production readiness is not a technical milestone —it is a risk acceptance decision.
Shipping a system means:

- accepting certain failures
- rejecting others
- committing to a specific worldview under uncertainty

What "Ready" Actually Means (AI-Native)

A system is *ready* when:

- the decision is clearly owned
- assumptions are explicit
- risks are documented
- failure modes are understood
- humans know when *not* to trust it

Readiness is about **clarity**, not confidence.

The Five Readiness Dimensions

A system must be reviewed across **all five** dimensions.

1️⃣ Decision Readiness

- Is the decision clearly defined?
- Is the cost of being wrong understood?
- Is reversibility documented?

If the decision is fuzzy, readiness is impossible.

2️⃣ Data & Assumption Readiness

- Are data sources defensible?
- Are label limitations known?
- Are key assumptions written down?

Hidden assumptions = hidden risk.

3️⃣ Pipeline Coherence

- Do cleaning, features, models, and metrics align?
- Are tradeoffs consistent end-to-end?
- Are earlier choices constraining later ones intentionally?

Local correctness is not enough.

4️⃣ Human Oversight Readiness

- Are review checkpoints defined?
- Are delegation boundaries explicit?
- Do humans know when to intervene?

Oversight cannot be “figured out later.”

5️⃣ Failure Awareness

- What does failure look like?
- How would it first appear?
- Who is responsible when it happens?

If failure cannot be described, it cannot be managed.

Common “False Ready” Signals

Be cautious if readiness is justified by:

- “metrics look good”
- “industry standard model”
- “we’ll iterate later”
- “we can monitor it”
- “it worked in a similar use case”

None of these address decision risk.

Readiness Is a Go / No-Go Gate

This module exists to give you language to say:

- “Yes, this is ready — and here’s why”
- “No, this is not ready — and here’s what’s missing”

Both are **success outcomes**.

🧪 Module 8 Mini-Exercise — Readiness Review

This is a **judgment exercise**, not a technical one.

Step 1 — Decision Summary

In 1–2 sentences:

- What decision does the system support?
- Why is deploying this decision risky?

Step 2 — Readiness Assessment

For **each dimension**, state whether the system is:

- ✅ Ready
- ⚠️ Partially Ready
- ❌ Not Ready

And explain *why* in one sentence each:

1. Decision readiness

2. Data & assumption readiness
3. Pipeline coherence
4. Human oversight readiness
5. Failure awareness

Step 3 — Go / No-Go Call

Make an explicit call:

Go / No-Go / Conditional Go

Briefly justify your decision.

Step 4 — Readiness Gap (If Not "Go")

If you chose anything other than **Go**, answer:

- What is the *single most important* gap preventing readiness?

Do **not** propose solutions — name the gap.

Rules (Important)

❌ No fixes
❌ No monitoring plans
❌ No retraining strategies
✅ Judgment only
✅ Risk ownership
✅ Explicit tradeoffs

Why This Module Closes DS Foundations

By completing this module, you demonstrate that you can:

- resist shipping pressure
- articulate readiness clearly
- protect stakeholders and users
- treat deployment as a **decision**, not a reward

This is the final skill before entering **Production Foundations**.

End-of-Curriculum Takeaway

The most senior move in data science is knowing when *not* to ship.

Relevant Articles

Checklist: Data Product Readiness

- https://www.toucantoco.com/en/blog/data-product-readiness-checklist?utm_source=chatgpt.com

Checklist for AI and ML Readiness

- https://passerelledata.com/checklist-for-ai-and-ml-readiness/?utm_source=chatgpt.com

How to Use Go No Go Decision Templates for Better Outcomes

- https://www.pageon.ai/blog/how-to-use-go-no-go-decision-templates-for-better-outcomes?utm_source=chatgpt.com

Technology Readiness Levels for Machine Learning Systems

- https://www.nature.com/articles/s41467-022-33128-9?utm_source=chatgpt.com